

Lab Assignment 2

Two-Level and Multi-Level Logic Optimization

Course: Synthesis and Optimization of VLSI Circuits

Assignment Title: Logic Optimization using Verilog HDL

Tools: Xilinx Vivado, Verilog HDL

Objective

The objective of this lab is to understand and implement **Two-Level and Multi-Level logic optimization** techniques using Verilog HDL and analyze their impact on **area and delay** after synthesis in Vivado.

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Instructions

1. Write synthesizable Verilog HDL code only.
 2. Do NOT use delays (#), `initial` blocks, or non-synthesizable constructs.
 3. Verify functionality using a testbench.
 4. Perform RTL synthesis in Vivado.
 5. Observe RTL schematic and synthesized design.
 6. Record LUT count and timing.
 7. Attach waveform screenshots and synthesis report.
 8. Compare results for original and optimized designs.
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Problem 1: Two-Level Logic Implementation

Task:

Implement the following Boolean function in **SOP form**:

$$F=ABC+AB'C+A'BC+A'B'C$$

Steps:

1. Write Verilog code using direct SOP expression.
2. Simulate and verify output.
3. Synthesize and note:
 - LUT utilization

- Logic depth
- 4. Minimize the function using K-map.
- 5. Rewrite optimized Verilog.
- 6. Compare synthesis results (before vs after).

Problem 2: POS vs SOP Realization

Task:

Implement the same Boolean function using:

- SOP form
- POS form

Compare:

- Area (LUTs)
- Delay

Problem 3: Multi-Level Logic Decomposition

Task:

Given:

$$F=AB+AC+BC$$

(a) Two-Level Version

Implement directly using AND–OR logic.

(b) Multi-Level Version

Factor the logic:

$$F=A(B+C)+BC$$

Write Verilog code using intermediate wires.

Compare:

- LUT count
- Critical path delay

Problem 4: Common Sub-expression Sharing

Task:

Implement:

$$\begin{aligned}F1 &= AB + C \\F2 &= AB + D\end{aligned}$$

1. First without sharing
2. Then with shared term AB

Observe hardware reduction.

Tools Required

- Xilinx Vivado
- Verilog HDL

Task Description

- Write Verilog code
- Simulate functionality
- Perform synthesis
- Analyze RTL and technology schematic
- Compare optimized vs unoptimized implementations

Expected Outcome

- Clear understanding of two-level vs multi-level logic
- Functional simulation
- Reduced hardware after optimization
- Ability to interpret synthesis reports

Evaluation Scheme (15 Marks)

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- Verilog coding correctness – 5
- Functional simulation results – 4
- RTL & synthesized schematic analysis – 3
- Viva / explanation – 2
- Quality of simulation and documentation-1

Learning Outcomes

Students will be able to:

- Implement two-level and multi-level logic
- Perform manual logic optimization
- Analyze synthesis results
- Understand logic factoring and sharing